

Deep Neural Networks - Mid-Semester 1 Shorts

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Last 30 minutes - DNN Midsem 1

Intuition

Use this route to recall, not to learn new theory: 5 minutes for perceptrons/MLPs, 5 for propagation, 5 for optimizers, 5 for generalization, 5 for CNNs/transfer, and 5 for mixed diagnosis.

Definitions and jargon

Say aloud: feature (input), label (target), parameter (learned number), affine map (weighted sum plus bias), activation (score transform), loss (error measure), logit (pre-probability score), gradient (loss slopes), regularization (constraint or penalty), dataset shift (train/deployment distributions differ), channel, receptive field, freeze, fine-tune.

Formulas, symbols, shapes, and assumptions

Recall: $z = w^T x + b$, where T turns a column into a row so $w^T x$ is a dot product; step 1 at $z \geq 0$; perceptron $w += \eta(y - \hat{y})x$, $b += \eta(y - \hat{y})$ with learning rate $\eta > 0$; a dense layer with n inputs and m outputs has $mn + m$ parameters; chain rule multiplies local slopes; Adam uses corrected running averages. CNN size is $\text{floor}((N + 2P - [D(K - 1) + 1])/S) + 1$ for input N , padding P , kernel K , stride S , dilation D ; floor rounds down.

Conditional properties

Perceptron convergence: finite and linearly separable data under standard repeated updates. Universal approximation: existence/capacity only. BN: training batch statistics, inference stored statistics; no universal shift cure. CNN formulas require valid positive stride/dilation and nonnegative numerator.

Important distinctions

Step nonlinear/boundary linear; bias affine/not linear; backprop computes gradients/optimizer updates; SGD sampling/momentum memory/adaptive scaling; training/validation/test; convolution convention/cross-correlation; freeze/fine-tune.

Checked example

For a $10 \times 10 \times 2$ input, 5 kernels of 3×3 , $P=1, S=1, D=1$, output is $10 \times 10 \times 5$; parameters are $3 \times 3 \times 2 \times 5 + 5 = 95$. State both shape and conditions.

Traps and counterexamples

No XOR convergence promise; no universal hidden-layer count for “deep”; no optimization/generalization promise from universal approximation; no global-convergence claim for deep optimizers; event cameras report brightness-change events, not frames; dilation belongs in CNN shape arithmetic.

Answer-free recall

Without notes, write the six core formulas, then answer: Why can training accuracy rise while test performance falls? When can transfer hurt? Which evidence separates a plateau from overfitting? What is the precise AND bias interval for $w = (1, 1)$?

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Session 1 - Neural-network fundamentals

Intuition

AI covers machines performing intelligent-seeming tasks. Machine learning (ML) learns patterns from data; deep learning (DL) is ML with multilayer neural networks. Supervised data have labels, unsupervised data do not, semi-supervised learning mixes both, and reinforcement learning uses actions and rewards.

Definitions and jargon

A feature is an input and a label the desired output. A weight scales a feature, a bias shifts the boundary, an activation transforms the score, and a loss measures quality. In a simplified analogy, dendrites receive signals, the axon hillock integrates, and the axon carries output; an artificial neuron is only mathematics. Connectionism builds behaviour from connected simple units.

Formulas, symbols, shapes, and assumptions

For length- d feature and weight columns x, w , $z = w^T x + b$; T turns a column into a row, so $w^T x$ sums paired products. Prediction $\hat{y} = f(z)$ is 1 when $z \geq 0$, else 0. Update $w \leftarrow w + \eta(y - \hat{y})x$, $b \leftarrow b + \eta(y - \hat{y})$, with learning rate $\eta > 0$. The error $y - \hat{y}$ is not a universal loss.

Conditional properties

A perceptron's boundary $w^T x + b = 0$ is linear although its step is nonlinear. Linearly separable means one line (or flat higher-dimensional boundary) separates the classes. Convergence needs finite, linearly separable data and repeated standard updates; XOR has no such promise.

Important distinctions

Label availability separates supervised, unsupervised, and semi-supervised learning; reinforcement learning uses action-reward interaction. Bias makes the score map affine, not linear. Training adjusts parameters; inference uses them.

Checked example

For AND with $w = (1, 1)$, the positive row $(1, 1)$ needs $2 + b \geq 0$, while $(1, 0)$ and $(0, 1)$ need $1 + b < 0$. Hence exactly $-2 \leq b < -1$; choose $b = -1.4$. The four scores are $-1.4, -0.4, -0.4, 0.6$, giving $0, 0, 0, 1$.

Traps and counterexamples

XOR's positive points are opposite corners, so one line cannot separate them. Not every AI uses ML or every DL task labels; bias is not linear, and artificial neurons are not biological equivalents.

Answer-free recall

Why does the threshold convention change the AND bias interval? Perform one perceptron update. Why does the nonlinear step leave one perceptron's boundary linear? [Know More](#) | [Detailed](#) | [External Sources](#)

Session 2 - Multilayer perceptrons

Intuition

One perceptron draws one boundary. An MLP composes many units so early layers create useful intermediate features and later layers combine them. Nonlinear activations prevent the entire stack from collapsing into one affine calculation.

Definitions and jargon

The input layer holds features, a hidden layer makes internal representations, and the output layer makes the prediction. Width is units in a layer; depth is successive computational transformations. State whether the non-computing input placeholder is counted. A dense layer connects every input to every unit. An affine map is a weighted linear map plus a bias.

Formulas, symbols, shapes, and assumptions

For a dense layer with input column x of shape $n \times 1$, weights W of shape $m \times n$, bias b of shape $m \times 1$, $z=Wx+b$ and $h=\phi(z)$, both $m \times 1$; ϕ is the activation. It has $mn+m$ trainable parameters. A $3 \rightarrow 5 \rightarrow 2$ MLP has $3*5+5 + 5*2+2 = 32$ parameters.

Conditional properties

With suitable nonlinear activation and enough units, universal-approximation results establish existence of parameters approximating certain functions, such as continuous functions on compact domains. They do not promise a small network, successful optimization, good test performance, or a required hidden-layer count.

Important distinctions

Depth and width are different. Representation capacity is different from trainability and generalization. Without nonlinear activations, composing $W_2(W_1x+b_1)+b_2$ gives one affine map $(W_2W_1)x+(W_2b_1+b_2)$.

Checked example

Build XOR with two hidden threshold units: $h_1=\text{OR}(x_1,x_2)$ and $h_2=\text{NAND}(x_1,x_2)$, then output $\text{AND}(h_1,h_2)$. For $(1,1)$, hidden values are $(1,0)$ and output is 0; for $(1,0)$, they are $(1,1)$ and output is 1.

Traps and counterexamples

Do not infer that deeper is always better, that every definition of DNN counts hidden layers identically, or that the universal-approximation theorem explains how gradient descent finds a solution.

Answer-free recall

Count weights and biases in a $4 \rightarrow 6 \rightarrow 3$ dense network. Explain why two affine layers without activation collapse. State three things universal approximation does not guarantee.

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Session 3 - Forward pass, backpropagation, and outputs

Intuition

A feedforward network sends values input-to-output without feedback. A computation graph records operations. Backpropagation walks it backward with the chain rule to compute loss gradients; gradient descent uses them.

Definitions and jargon

A logit is a score before probability conversion. A gradient collects loss slopes. The chain rule multiplies local slopes along a path. Sigmoid is $1/(1+\exp(-z))$; tanh is $(\exp(z)-\exp(-z))/(\exp(z)+\exp(-z))$ with range $(-1,1)$; ReLU is $\max(0,z)$. Softmax turns class logits into probabilities; $\exp$ is exponential and $\log$ its inverse.

Formulas, symbols, shapes, and assumptions

For K layers numbered $l=1,\dots,K$, layer l has n_l units, weights W_l , bias b_l , scores z_l , outputs h_l , and activation ϕ_l : $z_l=W_l h_{l-1}+b_l$, $h_l=\phi_l(z_l)$. If W_l is $n_l \times n_{l-1}$, the other three are $n_l \times 1$. Gradient descent is $\theta \leftarrow \theta - \eta dL/d\theta$: L is loss, θ all parameters, $\eta > 0$ the learning rate. Softmax is $p_k = \exp(a_k) / \sum_j \exp(a_j)$ over class logits; subtract the largest logit first.

Conditional properties

For one-hot targets, softmax cross-entropy is $L = -\log(p_y)$. Sigmoid and tanh can saturate and create small derivatives. Backpropagation calculates gradients; the optimizer updates.

Important distinctions

Forward propagation computes predictions/loss; backpropagation computes derivatives; gradient descent changes parameters. Softmax couples exclusive classes; independent labels normally use separate sigmoids.

Checked example

Let $x=3$, $z=2x-1=5$, $h=\text{ReLU}(z)=5$, and $L=(h-7)^2$. Then $dL/dx=2(5-7)(1)(2)=-8$. Separately, sigmoid at logit 1 is $1/(1+\exp(-1))=0.731$ (rounded).

Traps and counterexamples

Do not exponentiate huge logits naively. Sum-to-one does not prove calibration. Backpropagation is not an optimizer; cross-entropy is not perceptron error $y - \hat{y}$.

Answer-free recall

Trace shapes through a $5 \rightarrow 4 \rightarrow 3$ network. Explain the three stages: forward pass, backward pass, update. Why can subtracting the largest logit leave softmax probabilities unchanged?

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Session 4 - Optimizing deep models

Intuition

Deep loss is generally nonconvex. Plateaus are flat and slow; saddles curve up in some directions and down in others. Gradients guide updates, but no optimizer guarantees a global minimum.

Definitions and jargon

The gradient is the vector of loss slopes. Batch GD uses all examples, SGD one, and minibatch SGD a subset. Momentum smooths; Nesterov measures at a look-ahead. Rprop uses signs; safeguarded Quickprop uses a local quadratic step. Adagrad, RMSprop, and Adam use past squared gradients.

Formulas, symbols, shapes, and assumptions

At step t , θ_t is the parameter vector and g_t its gradient. Learning rate $\eta > 0$ sets scale; m_t, v_t average gradients and their squares; hats mean bias-corrected. Decays β_1, β_2 lie in $[0, 1)$; $\epsilon > 0$ prevents division by zero. Operations are elementwise.

1 Accumulate moments $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$ $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$ gradient average; squared-gradient average	2 Correct zero-start bias $\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$ $\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$	3 Update each parameter $\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$ products and divisions are elementwise
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Conditional properties

Bias correction fixes Adam's early zero-start bias. Adagrad's accumulator only grows, so steps can shrink. Rprop adapts by sign agreement; Quickprop needs denominator, growth, and sign safeguards.

Important distinctions

SGD chooses samples; momentum and adaptive scaling modify updates. Nesterov evaluates after look-ahead; epsilon stabilizes division, not the learning rate.

Checked example

With Adam $g_1=2$, $m_0=v_0=0$, $\beta_1=0.9$, $\beta_2=0.99$, $\eta=0.03$, epsilon negligible: $m_1=0.2$, $v_1=0.04$, $\hat{m}_1=2$, $\hat{v}_1=4$, so the first update is -0.03 .

Traps and counterexamples

No optimizer always wins. Large η can diverge; tiny η can stall. Quickprop needs caps for near-zero denominators and fallback on sign change.

Answer-free recall

Contrast batch, stochastic, and minibatch gradients. Compute a new first Adam step. Distinguish a saddle, local minimum, and plateau.

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Session 5 - Generalization and stable training

Intuition

Training loss measures fit to seen examples; the goal is performance on unseen data. Validation supports model choices; an untouched test set estimates final performance.

Definitions and jargon

Underfitting means train and validation results are poor; overfitting means train is strong while validation worsens. L1 adds absolute parameters; L2 their squares. Dropout masks training units. Dataset shift means train/deployment distributions differ. Internal covariate shift is the historical BN claim that layer inputs change as earlier layers learn—not dataset shift.

Formulas, symbols, shapes, and assumptions

$J(\theta) = \text{data_loss} + \lambda R(\theta)$: J is objective, θ parameters, R penalty, $\lambda \geq 0$ strength. L1 uses $\sum_i |\theta_i|$; L2 uses $\sum_i \theta_i^2$, summed over parameter index i . For inverted dropout, h is unmasked activation; mask m keeps entries with probability q , training scales by $1/q$, and inference uses h .

Conditional properties

Initialization can shrink/amplify signals and gradients. Xavier suits some symmetric activations; He scaling suits ReLU families. Gradient clipping caps value or norm to limit explosions, not vanishing. BN uses minibatch statistics in training and stored estimates at inference.

Important distinctions

L1 can encourage zeros; L2 smoothly discourages large weights. Dropout is random only in training. BN's historical motivation does not mean BN solves deployment shift.

Checked example

If unregularized loss is 0.42, $\lambda = 0.05$, and weights are (2, -1, 3), then L1 gives $0.42 + 0.05(6) = 0.72$; L2 gives $0.42 + 0.05(14) = 1.12$. These values compare objectives under the stated penalty definitions, not universal model quality.

Traps and counterexamples

Small training loss can coexist with poor validation. Repeated test use becomes tuning. BN cannot guarantee population robustness; dropout cannot repair bad labels.

Answer-free recall

Diagnose train-low/validation-high loss. Explain train, validation, and test roles. Compare L1, L2, dropout, initialization, and BN by the problem each addresses.

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Session 6 - Convolutional networks

Intuition

A convolutional layer scans a learned kernel across an image. Sharing that kernel preserves spatial structure and uses fewer parameters than a dense image-wide layer.

Definitions and jargon

This course uses cross-correlation: slide without flipping the kernel, multiply overlaps, and sum. Padding adds borders, stride controls movement, dilation spaces taps, channels hold feature maps, and pooling summarizes regions. Max pooling keeps the largest value; average pooling the mean. Kernels are learned.

Formulas, symbols, shapes, and assumptions

For input size N , padding P , kernel K , stride S , and dilation D , effective kernel $K_{\text{eff}}=D(K-1)+1$ and output size $\text{floor}((N+2P-K_{\text{eff}})/S)+1$; floor rounds down. Apply separately to height/width. With input/output channels $C_{\text{in}}, C_{\text{out}}$, parameters are $K_h \cdot K_w \cdot C_{\text{in}} \cdot C_{\text{out}} + C_{\text{out}}$ when each output channel has one bias.

Conditional properties

The formula needs integer settings, $S \geq 1$, $D \geq 1$, and nonnegative numerator. Pooling usually has no learned parameters. Padding preserves size only for compatible kernel, dilation, and stride.

Important distinctions

Parameters depend on kernels/channels, not image height/width. Stride skips positions; dilation spreads taps. Pooling aggregates; convolution learns. Mathematical convolution flips its kernel; cross-correlation does not.

Checked example

Input $15 \times 15 \times 3$, 4 kernels of size 3×3 , padding 2, stride 2, dilation 2: $K_{\text{eff}}=5$; output height and width are $\text{floor}((15+4-5)/2)+1=8$, so shape $8 \times 8 \times 4$. Parameters: $3 \cdot 3 \cdot 3 \cdot 4 + 4 = 112$.

Traps and counterexamples

Do not omit dilation from the output-size formula. “Same” padding is not automatically symmetric for every stride/kernel combination. A larger feature map does not itself increase one layer's parameter count.

Answer-free recall

Compute output shape and parameters for a fresh dilated layer. Explain why parameter sharing is useful. Distinguish padding, stride, dilation, channels, and pooling.

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Session 7 - CNN design and transfer learning

Intuition

CNNs grow receptive fields and build local patterns into task concepts. Residual connections add a direct identity path. Transfer learning reuses source-task features for applications such as image classification, detection, and medical imaging, subject to domain fit.

Definitions and jargon

The receptive field is the input region influencing one activation. A residual block adds learned change $F(x)$ to x . Freezing fixes reused parameters; fine-tuning updates them. Domain mismatch means source/target differences limit reuse.

Formulas, symbols, shapes, and assumptions

Residual output $y=F(x)+x$ needs matching shapes or a projection. Let r_l be receptive-field size and j_l input spacing between neighboring layer- l positions. With kernel K_l , dilation D_l , stride S_l :
 $r_l=r_{l-1}+D_l(K_l-1)j_{l-1}$ and $j_l=j_{l-1}S_l$.

Conditional properties

Freezing suits little target data or close domains; fine-tuning adapts but costs more and may overfit. Start with a frozen feature extractor and target-specific head, then unfreeze later blocks only if target validation supports it. Pretrained layers often use a small learning rate. AlexNet showed a large trained CNN; ResNet popularized residual learning.

Important distinctions

Focus on feature maps, downsampling, receptive fields, residual paths, and compute—not trivia. Severe domain mismatch can defeat transfer.

Checked example

Two stride-1 3×3 layers start with $r_0=1, j_0=1$. After layer 1, $r_1=3, j_1=1$; after layer 2, $r_2=5$. Thus one final activation can depend on a 5×5 input region.

Traps and counterexamples

Freezing can underfit a new domain; fine-tuning all can overfit tiny data. Residual addition needs matching shapes. Bigger is not automatically better.

Answer-free recall

Compute a receptive field through a stride-2 block. Choose freeze or fine-tune for a small related dataset and justify. Explain what a residual path changes and what it does not guarantee.

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Session 8 - Integrated diagnosis and review

Intuition

Session 8 adds no new theory. It asks you to connect symptoms to causes: representation limits, incorrect shapes, unstable optimization, poor generalization, or unsuitable CNN/transfer choices.

Definitions and jargon

A diagnosis links evidence to a likely mechanism and a targeted next check. Representation failure means the model cannot express the rule; optimization failure means training does not find useful parameters; generalization failure means training succeeds but unseen-data performance does not. A validity condition says when a formula applies; one valid counterexample disproves an absolute claim.

Formulas, symbols, shapes, and assumptions

Use a four-line answer frame: (1) name the governing idea or formula; (2) substitute values or trace shapes; (3) state conditions; (4) interpret and propose one targeted action. Keep the key formulas ready: perceptron update, dense parameter count, chain-rule product, Adam bias correction, regularized objective, and dilated CNN output size.

Conditional properties

The same symptom can have several causes. Flat training may reflect saturation, a plateau, unsuitable learning rate, or data problems; distinguish them with gradients, activation ranges, and train/validation curves. A recommendation is conditional on the evidence supplied.

Important distinctions

Training failure differs from generalization failure. A wrong shape differs from a wrong numerical value. Underfitting differs from optimization not yet converged. Dataset shift differs from overfitting, though both can hurt deployment performance.

Checked example

A model has training loss falling from 0.80 to 0.08, validation loss bottoming at 0.31 then rising to 0.57. This is evidence of overfitting after the validation minimum, not vanishing gradients. A targeted response is early stopping or stronger regularization, followed by validation; the untouched test set remains for final evaluation.

Traps and counterexamples

Do not recommend Adam for every problem, BN for every shift, more depth for every underfit, or transfer learning regardless of source-target mismatch. Do not diagnose from one metric without its split and trend.

Answer-free recall

Create one diagnosis each for XOR failure, exploding activations, widening train-validation gap, invalid residual addition, and negative CNN output numerator. For each, state one check and one bounded action.

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